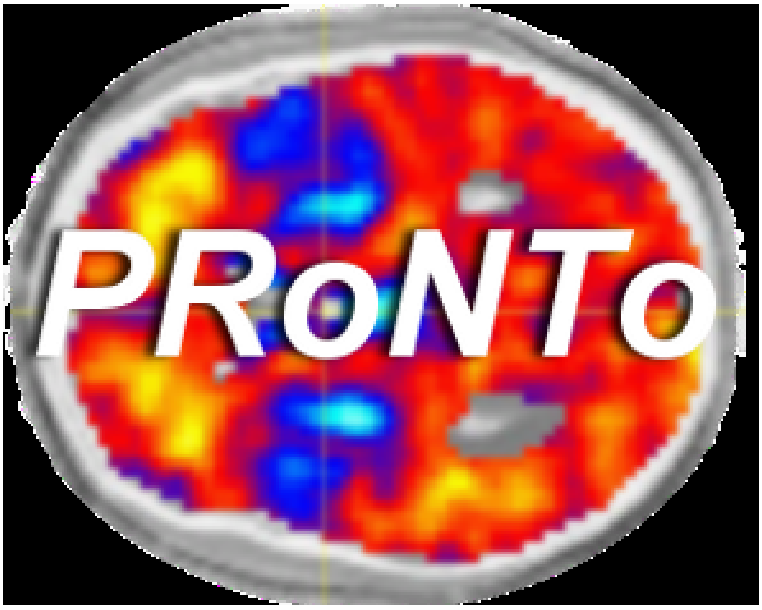

PRoNTo Manual

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Contents

Chapter 1

Introduction

This is where we introduce PRoNTo and explain what we wanted to do...
Lots of text to come!
Cite a few important references like [1], or [2], or another [3] and obvious [4].

Part I

Graphic User Interface

Chapter 2

Data & design

This is where we explain how data and design are defined within the GUI. Lots of text to come!

Chapter 3

Preprocessing

This is where we explain how preprocessing defined within the GUI.
Lots of text to come!

Chapter 4

Kernel computation

This is where we explain how kernel is computed within the GUI.
Lots of text to come!

Chapter 5

Cross validation

This is where we explain how cross-validation is performed within the GUI. Lots of text to come!

Chapter 6

Build model

This is where we explain how model is built within the GUI.
Lots of text to come!

Chapter 7

Apply model

This is where we explain how a model is applied within the GUI.
Lots of text to come!

Chapter 8

Review data

This is where we explain how you can review data within the GUI.
Lots of text to come!

Chapter 9

Review kernel & CV

This is where we explain how you can review the kernel and cross validation within the GUI. Lots of text to come!

Chapter 10

Display results

This is where we explain how you can display results within the GUI.
Lots of text to come!

Part II

Batching system

Chapter 11

Data & Design

Contents

Specify the group(s) of data set.

11.1 Directory

Select a directory where the PRT.mat file containing the specified design and data matrix will be written.

11.2 Groups

Add data and design for one group. Click 'new' or 'repeat' to add another group.

11.2.1 Group

Specify data and design for the group.

Name

Name of the group. Example: 'Controls'.

Select by

Depending on the type of data at hand, you may have many images (scans) per subject, such as a fMRI time series, or you may have many subjects with only one or a small number of images (scans) per subject , such as PET images. If you have many scans per subject select the option 'subjects'. If you have one scan for many subjects select the option 'scans'.

Subjects Add subjects.

Subject Add new modality for this subject.

Modality Add new data modality.

Name Name of modality. Example: 'BOLD'.

Interscan interval Specify interscan interval (TR). The units should be seconds.

Scans Select scans (images) for this modality. They must all have the same image dimensions, orientation, voxel size etc.

Data & Design Specify data and design.

Load SPM.mat Load design from SPM.mat (if you have previously specified the experimental design with SPM).

Specify design Specify design: scans (data), onsets and durations.

Units for design The onsets of events or blocks can be specified in either scans or seconds.

Conditions Specify conditions. You are allowed to combine both event- and epoch-related responses in the same model and/or regressor. Any number of condition (event or epoch) types can be specified. Epoch and event-related responses are modeled in exactly the same way by specifying their onsets [in terms of onset times] and their durations. Events are specified with a duration of 0. If you enter a single number for the durations it will be assumed that all trials conform to this duration.For factorial designs, one can later associate these experimental conditions with the appropriate levels of experimental factors.

Condition Specify condition: name, onsets and duration.

Name Name of condition.

Onsets Specify a vector of onset times for this condition type.

Durations Specify the event durations. Epoch and event-related responses are modeled in exactly the same way but by specifying their different durations. Events are specified with a duration of 0. If you enter a single number for the durations it will be assumed that all trials conform to this duration. If you have multiple different durations, then the number must match the number of onset times.

Regression targets (trials) Enter one regression target per trial.

Multiple conditions Select the *.mat file containing details of your multiple experimental conditions.

If you have multiple conditions then entering the details a condition at a time is very inefficient. This option can be used to load all the required information in one go. You will first need to create a *.mat file containing the relevant information.

This *.mat file must include the following cell arrays (each 1 x n): names, onsets and durations. eg. names=cell(1,5), onsets=cell(1,5), durations=cell(1,5), then names2='SSent-DSpeak', onsets2=[3 5 19 222], durations2=[0 0 0 0], contain the required details of the second condition. These cell arrays may be made available by your stimulus delivery program, eg. COGENT. The duration vectors can contain a single entry if the durations are identical for all events.

Time and Parametric effects can also be included. For time modulation include a cell array (1 x n) called tmod. It should have a have a single number in each cell. Unused cells may contain either a 0 or be left empty. The number specifies the order of time modulation from 0 = No Time Modulation to 6 = 6th Order Time Modulation. eg. tmod3 = 1, modulates the 3rd condition by a linear time effect.

For parametric modulation include a structure array, which is up to 1 x n in size, called pmod. n must be less than or equal to the number of cells in the names/onsets/durations cell arrays. The structure array pmod must have the fields: name, param and poly. Each of these fields is in turn a cell array to allow the inclusion of one or more parametric effects per column of the design. The field name must be a cell array containing strings. The field param is a cell array containing a vector of parameters. Remember each parameter must be the same length as its corresponding onsets vector. The field poly is a cell array (for consistency) with each cell containing a single number specifying the order of the polynomial expansion from 1 to 6.

Note that each condition is assigned its corresponding entry in the structure array (condition 1 parametric modulators are in pmod(1), condition 2 parametric modulators are in pmod(2), etc. Within a condition multiple parametric modulators are accessed via each fields cell arrays. So for condition 1, parametric modulator 1 would be defined in pmod(1).name1, pmod(1).param1, and pmod(1).poly1. A second parametric modulator for condition 1 would be defined as pmod(1).name2, pmod(1).param2 and pmod(1).poly2. If there was also a parametric modulator for condition 2, then remember the first modulator for that condition is in cell array 1: pmod(2).name1, pmod(2).param1, and pmod(2).poly1. If some, but not all conditions are parametrically modulated, then the non-modulated indices in the pmod structure can be left blank. For example, if conditions 1 and 3 but not condition 2 are modulated, then specify pmod(1) and pmod(3). Similarly, if conditions 1 and 2 are modulated but there are 3 conditions overall, it is only necessary for pmod to be a 1 x 2 structure array.

EXAMPLE:
Make an empty pmod structure:
pmod = struct('name',",'param',,'poly',);
Specify one parametric regressor for the first condition:
pmod(1).name1 = 'regressor1';
pmod(1).param1 = [1 2 4 5 6];
pmod(1).poly1 = 1;
Specify 2 parametric regressors for the second condition:
pmod(2).name1 = 'regressor2-1';
pmod(2).param1 = [1 3 5 7];
pmod(2).poly1 = 1;
pmod(2).name2 = 'regressor2-2';
pmod(2).param2 = [2 4 6 8 10];
pmod(2).poly2 = 1;

The parametric modulator should be mean corrected if appropriate. Unused structure entries should have all fields left empty.

Covariates Select a matrix/vector containing details of your covariates (i.e. any other data/information you would like to include in your design). If you enter a vector instead of a matrix, the entries of the vector will be repeated to create a matrix with the dimensions: number of scans x number of covariates (or vector entries.

No design Do not specify design. This option can be used for modalities (e.g. structural scans) that do not have an experimental design.

Scans Select this option if you have many subjects to spatially normalise, but there is one or a small number of scans for each subject.

Modality Specify modality, such as name and data.
Name Name of modality. Example: 'BOLD'.
Interscan interval Specify interscan interval (TR). The units should be seconds.
Subjects Select scans (images) for this modality. They must all have the same image dimensions, orientation, voxel size etc.

Regression targets (subjects)

Enter one regression target per subject. Enter matrix to specify targets for different modalities, [nsub x nmod]. The number of columns is the number of modalities. The number of rows is the number of subjects.

11.3 Masks

Select first-level (pre-processing) mask for each modality. The name of the modalities should be the same as the ones entered for subjects/scans.

11.3.1 Modality

Specify name of modality and file for each mask.

Name

Name of modality. Example: 'BOLD'.

File

Select one mask for each modality.

11.4 Review

Would you like to review the design?.

Chapter 12

Feature set / Kernel

Contents

Compute feature set according to the design specified

12.1 Load PRT.mat

Select data/design structure file (PRT.mat).

12.2 Name

Target filename for kernel matrix

12.3 Modalities

Add modalities

12.3.1 Modality

Specify modality, such as name and data.

Name

Name of modality. Example: 'BOLD'. Must match design specification

Scans / Conditions

Which task conditions do you want to include in the kernel matrix? Select conditions: select specific conditions from the timeseries. All conditions: include all conditions extracted from the timeseries. All scans: include all scans for each subject. This may be used for modalities with only one scan per subject (e.g. PET), if you want to include all scans from an fMRI timeseries (assumes you have not already detrended the timeseries and extracted task components)

All scans No design specified. This option can be used for modalities (e.g. structural scans) that do not have an experimental design or for an fMRI designwhere you want to include all scans in the timeseries

All Conditions Include all conditions in this kernel matrix

Voxels to include

Specify which voxels from the current modality you would like to include

All voxels Use all voxels in this modality

Specify mask file Select a mask for the selected modality.

Detrend

Type of temporal detrending to apply

None Do not detrend the data

Polynomial detrend Perform a voxel-wise polynomial detrend on the data (1 is linear detrend)

Order Enter the order for polynomial detrend (1 is linear detrend)

Discrete cosine transform Use a discrete cosine basis set to detrend the data.

Cutoff of high-pass filter (second) The default high-pass filter cutoff is 128 seconds (same as SPM)

Scale input scans

Do you want to scale the input scans to have a fixed mean (i.e. grand mean scaling)?

No scaling Do not scale the input scans

Specify from *.mat Specify a mat file containing the scaling parameters for each modality.

Chapter 13

Specify model

Contents

Construct model according to design specified

13.1 Data structure file

Select data/design structure file (PRT.mat).

13.2 Model name

Name for model

13.3 Use kernels

Are the data for this model in the form of kernels/basis functions? If 'No' is selected, it is assumed the data are in the form of feature matrices

13.4 Feature sets

Select feature sets to include in this model. These can be kernels or feature matrices. If multiple feature sets are included, they must all have the same number of rows

13.4.1 Feature set

Feature set to include in this model

Name

Name of a feature set

13.5 Model Type

Select which kind of predictive model is to be used.

13.5.1 Classification

Specify which elements belong to this class. Click 'new' or 'repeat' to add another class.

Class

Specify which groups, modalities, subjects and conditions should be included in this class

Name Name for this class, e.g. 'controls'

Groups Add one group to this class. Click 'new' or 'repeat' to add another group.

Group Specify data and design for the group.
Name Name of the group to include. Must exist in PRT.mat
Subject number(s) Subjects to be included in this class. Note that individual subject numbers (e.g. 1), or a range of subject numbers (e.g. 3:5) can be entered
Modalities Add modalities
MODALITY Specify modality, such as name and data.
Name Name of modality. Example: 'BOLD'. Must match design specification
Conditions / Scans Which task conditions do you want to include? Select conditions: select specific conditions from the timeseries. All conditions: include all conditions extracted from the timeseries. All scans: include all scans for each subject. This may be used for modalities with only one scan per subject (e.g. PET), if you want to include all scans from an fMRI timeseries (assumes you have not already detrended the timeseries and extracted task components)
Specify Conditions Specify the name of conditions to be included
Condition Specify condition:.
Name Name of condition to include.
All Conditions Include all conditions in this model
All scans No design specified. This option can be used for modalities (e.g. structural scans) that do not have an experimental design or for an fMRI designwhere you want to include all scans in the timeseries

13.5.2 Regression

Add one group to this regression model. Click 'new' or 'repeat' to add another group.

Group

Specify data and design for the group.

Name Name of the group to include. Must exist in PRT.mat

Subject number(s) Subjects to be included in this class. Note that individual subject numbers (e.g. 1), or a range of subject numbers (e.g. 3:5) can be entered

Conditions / Scans Which task conditions do you want to include? Select conditions: select specific conditions from the timeseries. All conditions: include all conditions extracted from the timeseries. All scans: include all scans for each subject. This may be used for modalities with only one scan per subject (e.g. PET), if you want to include all scans from an fMRI timeseries (assumes you have not already detrended the timeseries and extracted task components)

Specify Conditions Specify the name of conditions to be included
Condition Specify condition:.
NAME Name of condition to include.

All Conditions Include all conditions in this model

All scans No design specified. This option can be used for modalities (e.g. structural scans) that do not have an experimental design or for an fMRI designwhere you want to include all scans in the timeseries

Targets Specify continuous valued target variables

13.6 Machine

Choose a prediction machine for this model

13.6.1 SVM Classification

Binary support vector machine.

Arguments

Arguments for prt_machine_svm_bin.

13.6.2 Kernel Ridge Regression

Kernel Ridge Regression.

Regularization

Regularization for prt_machine_krr.

13.6.3 Custom machine

Choose another prediction machine

Function

Choose a function that will perform prediction.

Arguments

Arguments for prediction machine.

13.7 Cross-validation type

Choose the type of cross-validation to be used

13.7.1 Leave one subject out

Leave a single subject out each cross-validation iteration

13.7.2 Leave one subject per group out

Leave out a single subject from each group at a time. Appropriate for repeated measures or paired samples designs.

13.7.3 Leave one block out

Leave out a single block or event from each subject each iteration. Appropriate for single subject designs.

13.7.4 Custom

Load a cross-validation matrix. Note that an interface will be provided for this functionality in a later release

13.8 Data Operations

This branch controls operations that can be applied to the data before the data is passed to the classifier. Add zero or more operations to be applied. These will be executed in the order specified.

13.8.1 No operations

No design specified. This option can be used for modalities (e.g. structural scans) that do not have an experimental design or for an fMRI designwhere you want to include all scans in the timeseries

13.8.2 Specify operations

Specify operations to apply

Operations

Add zero or more operations to be applied to the data. These will be executed in the order specified.

Operation Select an operation to apply.

Chapter 14

Run model

Contents

Trains and tests the predictive machine using the cross-validtion structure specified by the model.

14.1 Load PRT.mat

Select PRT.mat (file containing data/design structure).

14.2 Model name

Name of a feature set

Part III

Data processing examples

Chapter 15

Data set 1

This is where we explain how how to process data set 1.

Lots of text to come!

Chapter 16

Data set 2

This is where we explain how how to process data set 2.

Lots of text to come!

Part IV

Advanced topics

Chapter 17

PRT structure

This is where we explain how the PRT structure is organised.

Lots of text to come!

Chapter 18

List of PRoNTo functions

This is where we list the PRoNTo functions.

Lots of text to come!

Bibliography

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